Manufacturing Analytics Project Report

Tools: **Excel, SQL, Power BI, Tableau**

Project Code: **P1202**

Group Number: **2**

1. Introduction

Manufacturing industries generate a large volume of operational data related to production, processing, rejection, and revenue.

Analyzing this data helps organizations understand operational efficiency and improve decision-making.

This project focuses on analyzing a manufacturing dataset using SQL and Business Intelligence tools to derive meaningful insights.

The analysis helps evaluate production performance, identify wastage and rejection patterns, and understand revenue generation across departments and machines.

2. Objectives of the Project

- Analyze manufacturing production data and measure operational performance.
- Calculate total processed, produced, and rejected quantities.
- Identify departments and machines contributing the most to production.
- Evaluate wastage generated during the manufacturing process.
- Analyze revenue generated by buyers and departments.
- Visualize manufacturing performance through dashboards.

3. Tools and Technologies Used

SQL (MySQL) – Data storage and analysis

Excel – Data preparation and preprocessing

Power BI / Tableau – Dashboard creation and visualization

Manufacturing Dataset – Source of production data

4. Dataset Description

The dataset contains manufacturing production records including buyer information, department details,

employee details, machine codes, production quantities, rejected quantities, and revenue values.

Important fields include Buyer, Customer Name, Department Name, Employee Name, Machine Code, Produced Quantity,

Processed Quantity, Rejected Quantity, Total Value, and Production Date.

5. Data Analysis Performed

The dataset was analyzed in multiple stages including Overview Analysis, Production Analysis, Wastage and Rejection Analysis, and Revenue Analysis. SQL queries were used to calculate total production, processed quantities, rejected quantities, and wastage. Machine-wise, department-wise, and employee-wise performance were also evaluated.

6. Dashboard Visualization

The processed data was visualized using Power BI/Tableau dashboards including production KPIs, department-wise production charts, monthly production trends, wastage analysis, and revenue performance visuals. These dashboards allow stakeholders to quickly monitor manufacturing performance.

7. Key Insights

- Certain departments contribute more to production output.
- Some machines handle a majority of manufacturing operations.
- Rejection rates vary across departments and machines.
- Wastage occurs due to differences between work order quantity and produced quantity.
- Production and revenue trends vary across months.

8. Conclusion

This project demonstrates how data analytics can improve manufacturing operations.
Using SQL queries and BI dashboards, raw manufacturing data was transformed into meaningful insights related to production, wastage, and revenue. The analysis supports better operational decisions and helps improve efficiency within manufacturing processes.